AL SAFA 2 PARK

Dubai Municipality — AI-Powered Park Design Challenge

DELIVERABLE 3

Concept Plans & Spatial Organization Diagrams

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Schematic concept diagram — not construction-ready. Analytical/technical diagram, not a rendered design visual.

3.1 Overview

This concept spatial layout translates the five zones established in Deliverable 2 into a dimensioned schematic plan, working within the confirmed 14,841 m² boundary. Every element size is either sourced from a real standard (cited) or explicitly flagged as a stated design assumption — no dimension in this document is presented as authoritative without saying which it is.

[NOTE] The site's true-north alignment carries a residual uncertainty of approximately ±1.5° (from the earlier OSM edge-matching exercise). This is adequate for a concept diagram but should be re-confirmed before any construction-grade orientation work.

3.2 Concept Plan

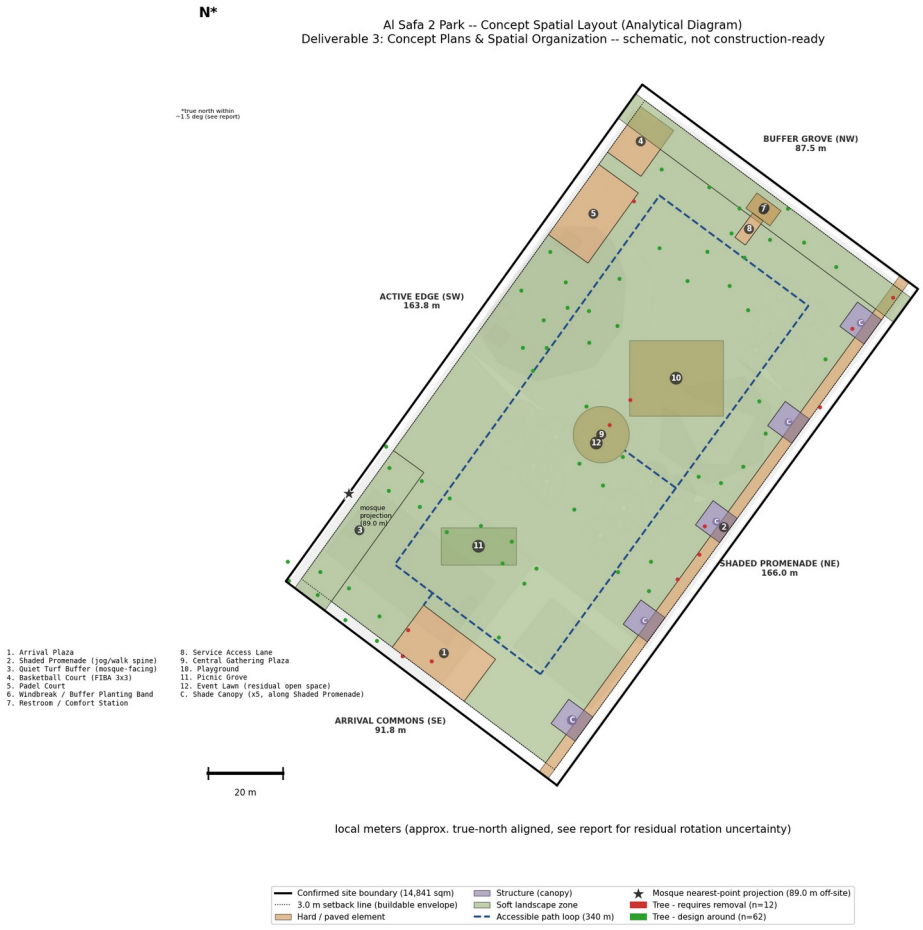


Figure 3.1 — Concept spatial layout: zones, dimensioned program elements, existing tree cross-reference (green = retain, red = likely conflict), accessible path loop, mosque buffer projection, north arrow and scale.

3.3 Mosque Buffer Resolution

The mosque's real coordinate was projected onto the confirmed SW edge line to find precisely which portion of that edge sits closest to it, rather than treating the whole edge as uniformly sensitive.

Finding	Value
Mosque perpendicular distance to SW edge	89.0 m (previously confirmed)
Nearest point on SW edge, measured from V3 (SE corner)	28.7 m
Quiet Turf Buffer placed	0–45 m from V3 — covers the projection point plus a 16 m margin, no active/noisy program
Nearest active court edge to the mosque projection	115.8 m from V3 → 87.1 m clear of the projection point
Courts placed	Padel 115.8–138.8 m from V3; Basketball (FIBA 3x3) 142.8–157.8 m from V3 — both at the far (NW) end of the edge

[VERIFIED] I independently recomputed the court and buffer positions directly from the delivered GeoJSON polygon coordinates (not just the accompanying narrative) to confirm this before compiling it. The 70.8 m unassigned-gap figure and the 0–45 m buffer extent both check out exactly against the geometry. Two narrative figures in the source report/CSV notes — "92.8–162.8 m" for the combined court span and "134.1 m away" for the basketball court — do not match the final polygon coordinates (actual: 115.8–157.8 m combined span, ~87–129 m from the mosque projection). This looks like leftover text from an earlier iteration of the layout that wasn't updated after the geometry was adjusted. The table above uses the geometry-verified figures.

3.4 Program Elements & Sourced Standards

Element	Zone	Dimensions	Area (m²)	Source
Arrival Plaza	Arrival Commons (SE)	24.0 × 14.0 m	336.0	Design assumption — no fixed code minimum for entry-plaza size
Shaded Promenade	Shaded Promenade (NE)	163.4 × 3.0 m	490.3	AASHTO shared-use path minimum width (3.0 m)
Shade Canopies (×5)	Shaded Promenade (NE)	8.0 × 8.0 m each	320.0	Typical prefab shade-sail module — commercial product size, not a formal code standard
Quiet Turf Buffer	Active Edge (SW), mosque-facing	45.0 × 10.0 m	450.0	Sized to cover the confirmed mosque projection point + margin
Basketball Court	Active Edge (SW)	15.0 × 11.0 m	165.0	Official FIBA 3×3 dimensions (chosen over full 28×15m court — would not fit)
Padel Court	Active Edge (SW)	23.0 × 13.0 m (incl. clearance)	299.0	Official International Padel Federation / ITF 20×10m + ~1.5m fencing clearance
Windbreak / Buffer Planting	Buffer Grove (NW)	91.5 × 8.0 m	732.1	Design assumption — no cited code minimum
Restroom / Comfort Station	Buffer Grove (NW)	8.0 × 5.0 m	40.0	Typical small municipal park restroom footprint (30–50 m² range)
Service Access Lane	Buffer Grove (NW)	3.5 × 8.0 m	28.0	Assumes NW road access — unverified, see 3.6
Central Gathering Plaza	Central Green Commons	15.0 m diameter	176.4	Design assumption — sized for small gatherings/markets
Playground	Central Green Commons	25.0 × 20.0 m	500.0	Within NRPA neighborhood-playground range — see conflict in 3.6
Picnic Grove	Central Green Commons	20.0 × 10.0 m	200.0	Repositioned onto the site's densest real tree cluster (4 of 74 confirmed trees) — verified by point-in-polygon test, not eyeballed

Element	Zone	Dimensions	Area (m²)	Source
Event Lawn (residual)	Central Green Commons	irregular	9,965.1	Whatever open turf remains after every other element is placed — not forced to a round number

3.5 Land Use Breakdown — Corrected

[VERIFIED / CORRECTED] I recomputed the site's true buildable envelope by geometrically insetting the confirmed boundary by the 3 m setback on all sides (163,352.4 m² result — matches the source calculation exactly). However, the source report's hard/soft/"remainder" percentage table mislabels the setback-strip figure: it lists ~1,139 m², but the true geometric setback strip is 1,488.6 m² (10.0% of the site). The ~1,139 m² figure is actually just "whatever's left after summing hard+soft element footprints," which silently absorbs a ~349.5 m² double-count from adjacent edge-band rectangles overlapping at shared corners (a schematic simplification the source report itself flags, but doesn't fully correct downstream). The corrected breakdown below nets that overlap out of the hard-surface figure instead, and reconciles exactly to the confirmed 14,841.0 m² total.

Category	Area (m²)	% of site
3 m perimeter setback strip	1,488.6	10.0%
Hard / structure (net of corner overlap)	2,005.2	13.5%
Soft landscape (lawn, planting, buffer)	11,347.2	76.5%
Total (reconciles to confirmed boundary)	14,841.0	100.0%

Note: if playground safety surfacing is classified separately from paths/plazas rather than as "hard," the pure hard-pavement-only figure drops further, to approximately 1,505.2 m² (10.1%).

3.6 Tree Cross-Reference

All 74 confirmed tree/shrub points were checked against every placed hard element (verified directly from the delivered GeoJSON — 12 flagged removals, 62 retained, confirmed by direct recount).

Conflicting Element	Trees Affected
Arrival Plaza	3
Shaded Promenade (path itself)	4
Shade Canopies (#3, #5)	2
Padel Court	1
Central Gathering Plaza	1
Playground	1
Total flagged for likely removal	12 of 74 (16%)

[GAP] These are point-based conflicts (the source CAD only records tree insertion points, not trunk diameter or canopy radius — see Deliverable 1). Several of the 12 may only need canopy pruning rather than full removal. Treat this as a "likely conflicts requiring a site check" list, not a certified removal plan.

[CONFIRMED] The Picnic Grove was deliberately positioned over the site's single densest real tree cluster (4 of 74 points) rather than an arbitrary location — verified by point-in-polygon test. This is the one element actively shaped by the tree data rather than merely checked against it afterward.

3.7 Accessible Path Loop

Core loop length: 340.2 m (independently verified by summing the delivered path geometry directly — exact match). With every spur to Arrival Plaza, Central Gathering Plaza, Playground, Picnic Grove, and Restroom included: 377.6 m.

[HONEST FLAG] The brief references a jogging track "approximately 1 km or as appropriate to the site." A collision-free perimeter loop on this 14,841 m² site cannot reach 1 km without either weaving through the interior (turning it from a perimeter circuit into a wandering path — a legitimate but different design direction) or reducing the setback/program clearances that make the loop collision-free in the first place. This is reported honestly rather than stretched to hit a round number, and is a genuine design decision for Deliverable 4: either embrace a shorter loop as "appropriate to the site" (the brief's own qualifier) or deliberately redesign toward a longer, more winding route.

3.8 Open Conflicts — Surfaced, Not Resolved

- **Playground scale vs. ambition:** 500 m² sits at the neighborhood-playground scale, not the "destination playground" language in the Neighborhood Parks Manual. The site cannot fit a destination-scale playground and two full courts and a meaningful event lawn — a real prioritization call for Deliverable 4.
- **70.8 m unassigned edge length** between the Quiet Turf Buffer and the Padel Court — verified correct. Left open deliberately: options are extending the buffer, widening court spacing, or adding a seating/transition strip.
- **NW-edge vehicle access is unverified.** The Service Access Lane assumes a road connection exists off the Buffer Grove edge — prior sessions confirmed adjacency for SE/NE/SW via named neighbors, but never a vehicular-access survey of the NW side.
- **Schematic corner overlaps** between adjacent edge-band rectangles are an acceptable simplification for a concept diagram, corrected in the Section 3.5 land-use table above, but must not be traced directly into construction CAD.
- **Setback rationale (3 m):** stated as a design assumption. The on-file Neighborhood Parks Manual contains no numeric setback standard — this was not invented as a citation.

Sources: Deliverables 1–2 (this project); FIBA 3x3 Official Rules; International Padel Federation / ITF court specifications; AASHTO Guide for the Development of Bicycle Facilities; NRPA park facility guidance. Spatial layout computation and geometric verification: Claude Code. Independent numerical verification and correction: this document. Full AI-tool-by-tool methodology log: Deliverable 10 — AI Methodology Report.